

Problem Set 9

1. We have previously looked at a few different variations of knockout swaps. In Chapter 6 we examined a period-by-period knockout swap in which each period of the trade can be knocked out irrespective of what happens in other periods. In Problem Set 6, we considered a full knockout swap that was a 5-year trade that could be knocked out after a year, depending on where the 4-year swap rate was. Notice that this example considered the special case in which the reference rate (which was the 4-year swap rate prevailing in a year) was the same as the remaining maturity of the trade at the time at which the trade could be knocked out.

Suppose the swap curve is flat at 7% and consider the following trade:

Swaps 'R Us pays:	Fixed
Client pays:	1-year LIBOR
Notional:	\$100mm
Maturity:	5 years
Payment Frequency:	Annual, on both sides
Day Count:	30/360, on both sides
Knockout Feature:	If, in one year, 1-year LIBOR is less than or equal to the Knockout Trigger, then there will be no further payments in the swap on either side
Knockout Trigger:	5.75%

a) Let's consider two cases. In the first case, the 4-year swap rate, one year forward and 1-year LIBOR, one year forward are positively correlated. In the second case, the 4-year swap rate, one year forward and 1-year LIBOR, one year forward are negatively correlated. In which of these two cases will the fixed rate that Swaps 'R Us is willing to pay in this trade be higher?

b) Suppose that we had a model to price up this trade that made use as a pricing parameter the correlation between the 4-year swap rate, one year forward and 1-year LIBOR, one year forward. Is it possible that this pricing parameter be set to zero in our model?

2. Suppose that the 2-year swap rate (quoted semi bond versus 3-month LIBOR quarterly money) is currently 6%. Consider a 3 month into 2-year receiver swaption struck at 5.70% that ceases to exist if the 2-year swap rate trades at or below 5.60% during a "knockout period." That is, the buyer of this 3 month into 2-year receiver swaption will not get a payoff at expiry in

three months if the 2-year swap rate trades at or below 5.60% during the knockout period.

Consider three different potential knockout periods for this trade:

- European: one time in exactly one month from today.
- Bermudan: two times, in exactly one month from today and again in exactly two months from today.
- American: any time between one month from today and two months from today.

a) List these options with the various knockout periods from most expensive to least expensive.

b) Suppose that we buy the European version of this knockout option. Suppose that immediately after we buy the option, the correlation between the 2-year swap rate, 1 month forward and the 2-year swap rate, 3 months forward decreases. However, despite this change in correlation, assume further that swap rates remain the same and 3 month into 2-year swaption vol remains the same. Given the move in correlation, what happens to the value of the European knockout option?

3. Consider again our stylized example in which swaps pay annual bond on the fixed side versus 1-year LIBOR annual bond. Suppose that a dealer underwrites a non-inversion note and in conjunction with this note the dealer enters into the following swap transaction:

Notional:	\$100 mm
Maturity:	10 Years
Dealer pays:	Year 1: 6%
	Years 2-10: $6\% \times N/M$
	Where N = number of days in accrual period that
	Curve ≥ 0
	M = number of days in accrual period
	Curve = 10-year CMS 2-year CMS
Issuer pays:	1-year LIBOR
Resets:	Daily for 10-year CMS and 2-year CMS for
	purposes of calculating the number of days the
	coupon is accrued; annual for the purposes of
	1-year LIBOR settings
Payment Frequency:	Annual, on both sides
Day Count:	30/360, on both sides

Suppose that at the time this trade is executed, the 2-year swap rate is 2.25% and the 10-year swap rate is 4.50%. Thus, 2s10s on a spot basis is 225 bps. Further assume that on a forward basis, the 2s10s curve is at no point over the next ten years flatter than 25 bps.

When the swap is first executed, the fixed coupon is set such that the trade is an $NPV = 0$ transaction. Assume that when pricing the trade, the dealer uses a model that employs a “flat” (i.e., non-time varying) correlation parameter between 10-year CMS and 2-year CMS.

Which of the following is true:

- Immediately after executing the swap, the dealer is *long implied correlation* (meaning that an increase in implied correlation increases the value of the swap to the dealer).
- Immediately after executing the swap, the dealer is *short implied correlation* (meaning that a decrease in implied correlation increases the value of the swap to the dealer).
- There is not enough information to determine whether the dealer is long or short implied correlation.

4. Consider the pricing of a receiver swaption on a forward starting swap. In particular, consider a 1 into 2x5 struck at 4%. This is an option that expires in one year and gives the holder the right to receive fixed at 4% in a 3-year swap, effective two years after expiry.

We could certainly consider pricing this swaption regarding the underlying forward starting swap as the underlying, make distributional assumptions about this forward swap rate, and employ a model to price up the option.

Alternatively, recall that a forward starting swap can be decomposed into a position in two spot starting swaps. If we were to regard this trade as an option on a portfolio of these two spot starting swaps, we would need to consider the correlation of the two spot starting swaps in order to price the option.

Suppose that 1 into 2 swaption vol, 1 into 5 swaption vol, and swap rates are unchanged, but the correlation between the 1x3 swap rate (i.e., the 2-year swap rate, one year forward) and the 1x6 swap rate (i.e., the 5-year swap rate, one year forward) increases. Which of the following is true?

- The value of the option increases.
- The value of the option decreases.
- The value of the option remains the same.
- There is not enough information provided to determine the answer.

5. Suppose an investor enters into a forward vol trade. In particular, suppose the investor buys 5 into 10 swaptions, five years forward. Unless otherwise specified, suppose that the contract is settled on the basis of the then-prevailing price of the underlying swaption straddle at the settlement of the contract. Assume we use the normal model to price plain vanilla swaptions. For each of the questions below, answer with one of the following:

- The value of the forward vol contract increases from the perspective of the investor.
- The value of the forward vol contract decreases from the perspective of the investor.
- The value of the forward vol contract remains the same.
- There is not enough information provided to determine the answer.

a) How does the value of the contract change if there is a parallel shift upwards of the entire normal implied swaption matrix immediately after the trade is executed? Assume that this parallel shift upwards is on an absolute basis, not a relative basis (i.e., all normal implied swaption vols increase by the same number of basis points, not by the same percentage). In addition, assume that swap rates do not move given the change in vol.

b) How does the value of the contract change if 10 into 10 normal implied swaption vol increases immediately after the trade is executed? Assume that swap rates do not move given this change in vol.

c) How does the value of the contract change if 10 into 10 normal implied swaption vol increases immediately after the trade is executed? Assume that swap rates do not move given this change in vol and assume that no other implied swaption vol changes other than 10 into 10 implied swaption vol.

d) How does the value of the contract change if 5 into 10 normal implied swaption vol increases immediately after the trade is executed? Assume that swap rates do not move given this change in vol.

e) How does the value of the contract change if swap rates decline (assume a parallel shift in rates) immediately after the trade? Assume that normal implied swaption vols do not change given this move in rates.

f) For this question, assume that the forward vol contract is settled on the basis of the then-prevailing normal implied vol of the underlying swaption straddle. How does the value of the contract change if swap rates decline (assume a parallel shift in rates) immediately after the trade? Assume that implied swaption vols do not change given this move in rates.

Appendix A

Refresher in Option Pricing

Many things in this appendix underly the options-related material covered throughout the text. If you are not familiar with the material discussed here, I suggest you become intimately familiar with it. This appendix provides a brief review of the fundamentals of option pricing. We'll begin with some basic definitions and concepts and establish boundary conditions that option prices must obey. We'll review the binomial option pricing model and introduce a concept known as risk-neutral valuation, a major paradigm in the pricing of options. We'll review the basics of the Black-Scholes model and examine several quantities that are commonly used to measure how option prices change when other variables change. We will discuss some exotic options and the concept of a package. Again, this material is intended to be both brief and a review. If you need more detail, I refer you to Hull (2008) for a more complete treatment.

A.1 The Basics

We will make the following assumptions throughout:

- There are no arbitrage opportunities.
- Markets are frictionless (i.e., there are no transaction costs).
- All individuals can borrow or lend at the same risk-free rate, r , which is assumed to be positive and constant through time.

Trading Strategy	Cash Flow Today	Cash Flow in the Future	
		Heads	Tails
A	\$0	\$1	\$0
B	\$1	\$0	\$0
C	−\$0.05	\$1	\$1
D	\$0	\$0	\$0

Table A.1: Some Sample Trading Strategies

An *arbitrage* is a trading strategy that enables us to lock in a riskless profit. To be a bit more formal, an arbitrage is a strategy in which we have nonnegative cash flows today, nonnegative cash flows in all states of nature in the future, and strictly positive cash flow either today or in at least one state of nature in the future.

To gain some intuition behind this definition, suppose that there are two states of nature in the future, called heads and tails, and consider the independent trading strategies in Table A.1. Which of these four strategies represent an arbitrage opportunity?¹

Notice that under the assumption of no arbitrage, if two assets have exactly the same payoff in each state of the world in the future, then they must have the same price today. If they did not have the same price, then we could buy the cheap one and sell the expensive one, resulting in positive cash flow today. Our cash flows in the future would be zero, as any cash flow emanating from the asset we are short would be exactly offset by our

¹Strategies A and B represent arbitrage opportunities, since each strategy has nonnegative cash flows both today and at all states of nature in the future, and each has strictly positive cash flows either today or in some state of nature in the future. Strategy C, although it looks like a “good deal,” is strictly speaking not an arbitrage since it has a negative cash flow today. It could be the case that Strategy C lends itself to an arbitrage. For example, suppose that one is able to borrow \$0.05 today with a promise to pay back \$0.06 in either state in the future. If this is the case, then combining Strategy C with this loan we can form a trading strategy, call it C', consisting of zero cash flow today and a payout of \$0.94 in each state in the future, which is surely an arbitrage. So given the fact that Strategy C is so seemingly an arbitrage itself, why don't we just call it one? There is no telling how long the time difference is between now and the future, or how high interest rates are. For instance, suppose that borrowing \$0.05 today necessitated an investor to pay back \$1.10 in the future. In this case, not only is Strategy C not an arbitrage—it is a really bad deal. Strategy D is not an arbitrage. Although it has nonnegative cash flows both today and at all states of nature in the future, it does not have strictly positive cash flows either today or in at least one state in the future.

long. This strategy would yield an arbitrage, since there would be positive cash flows today and no cash flows in the future. Thus, in order to avoid arbitrage, it must be that the two assets have the same price today.

How reasonable are these assumptions? It depends. It's true that arbitrage opportunities come about on occasion, but most financial markets are such that any systematic arbitrages would never last long before prices moved back in line. The assumption of zero transaction costs is not very reasonable since most market participants pay bid/offer when they want to transact, but we'll make this assumption for simplicity.² The third assumption of nondifferentiated borrowing and lending rates is not terribly realistic. In fact, during the financial crisis that first manifested itself in 2007, certain institutions were having a very difficult time borrowing at all. The seemingly innocuous assumption that the risk-free rate be positive even became suspect in the aftermath of the financial crisis, when the yields on certain short-term U.S. Treasury bills were negative.³ Nonetheless, we will go ahead with these assumptions, bearing in mind their shortcomings.

A *derivative* or *derivative security* is a contract or asset whose value depends on some underlying asset. An *option* is a particular type of derivative. All options are derivatives, but not all derivatives are options.⁴ There are two types of basic options. The owner of a *call option* has the right, but not the obligation, to buy an underlying asset within some prespecified time at some prespecified price. The owner of a *put option* has the right, but not the obligation, to sell an underlying asset within some prespecified time at some prespecified price. The prespecified price is known as the *strike* or *exercise* price. If the owner of the option may exercise only at maturity, the option is said to be *European*. If exercisable at any time at or before maturity, the option is said to be *American*.

The variables that determine the value of an option include the current underlying asset price S , the strike price X , the time remaining to expiration T , the volatility of the underlying asset σ , the risk-free rate of interest r , and any dividends d paid by the underlying asset during the life of the option.⁵

²For the pricing of options with transaction costs, see, for example, Leland (1985).

³The negative yields on these bills were a function of the high demand for these securities. As investor demand for short-dated bills kept pushing prices up and eventually above par, the yields on these bills went negative. The interpretation that investors were willing to take a small loss for the certainty of getting most of their money back has some merits. It is important to note that for many interest rate derivatives the risk-free rate employed in option pricing is not based on Treasury rates but rather on LIBOR rates. As discussed in Chapters 1 and 3, LIBOR is an important benchmark in the swaps market.

⁴A futures contract, for example, is a derivative that is not an option.

⁵In some cases, the risk-free rate will be quoted assuming continuous compounding, and at other times it will be quoted assuming simple compounding. Let r_c and r denote the risk-free rate quoted assuming continuous compounding and simple compounding, respectively. The present value of a dollar received in a year must be the same, regardless

C	American call option
c	European call option
P	American put option
p	European put option
r	the risk-free rate
S_T	asset price at expiration
S	current underlying asset price
σ	volatility of the underlying asset
d	dividends
T	time to expiration of option
X	exercise price

Table A.2: Some Symbols Used in This Appendix

This and other notations are summarized in Table A.2. Unless stated otherwise, we will assume that the underlying asset does not pay any dividends during the life of the option.

The value of an option is a function of these underlying variables. For example, in the case of a call option whose value today is denoted c , it would be most appropriate to represent this as $c(S, X, T, \sigma, r, d)$. This is, however, a lot of notation to carry around, and most often we will just write c and bear in mind the dependence of c on these variables. Other times, in order to highlight the dependence of c on one or more of these variables in particular, we may employ notation such as $c(X)$, in this case indicating the dependence of c on the strike X . However, it is important to remember that despite highlighting the dependency of the option price on one particular variable, all these variables drive the price of the option.

As discussed in Hull (2008), the effect on option prices of an increase in any of these variables (while holding all other variables fixed) is shown in Table A.3. Most of these results are pretty straightforward. We will skip over most of them but will highlight a few that are not entirely obvious:

- An increase in volatility increases both call and put prices. This result follows from the limited downside the holder of a call or a put has. An increase in volatility increases the spread of the distribution of the ter-

of the method of quoting interest. Thus, we must have

$$e^{-r_c} = \frac{1}{1+r},$$

or $r_c = \ln(1+r)$.

Variable Increased	Effect on Option Price			
	European Call	American Call	European Put	American Put
Asset Price	↑	↑	↓	↓
Strike Price	↓	↓	↑	↑
Time to Expiry	?	↑	?	↑
Riskless Rate	↑	↑	↓	↓
Volatility	↑	↑	↑	↑
Dividends	↓	↓	↑	↑

Table A.3: Effect of Changing Variables that Determine Option Prices

minal asset price. Thus, there is a greater chance the option will finish deeper *in-the-money*, meaning that the underlying asset price will be above (below) the strike price for a call (put). Although an increase in volatility also means that there is a greater chance that the option will finish deeper *out-of-the-money*, meaning that the underlying asset price is below (above) the strike price for a call (put), the holder of the option can lose at most his premium.

- Options always increase in value as time to expiration increases, right? Wrong. This is true for American options but is not always true for European options. Consider two American options on the same underlying with the same strike price. One expires at time t_1 and the other at time t_2 where $t_1 < t_2$. If the American option that expires at t_2 ever traded at a price less than the American that expires at t_1 , we could buy the long-dated option and sell the short-dated option, pocket money today, and exercise the long-dated option to cover our short if the short were ever exercised against us.⁶ But then this would be an arbitrage opportunity. Inasmuch as we assume there are no arbitrage opportunities, it must then be that the longer-dated American is worth more than the shorter-dated American.

This argument doesn't hold for Europeans, however. In fact, sometimes European options with shorter times to expiry are worth more

⁶Note that this is not to say that exercising our long-dated American call option if the short-dated American call is exercised is our optimal strategy. All we are saying is it's enough to cover our short, guaranteeing nonnegative cash flows at all times in the future. Indeed, if the underlying asset pays no dividends, it is definitely not in our best interest to exercise the American option early, as we will soon see.